

Quantum Technology and Artificial Intelligence

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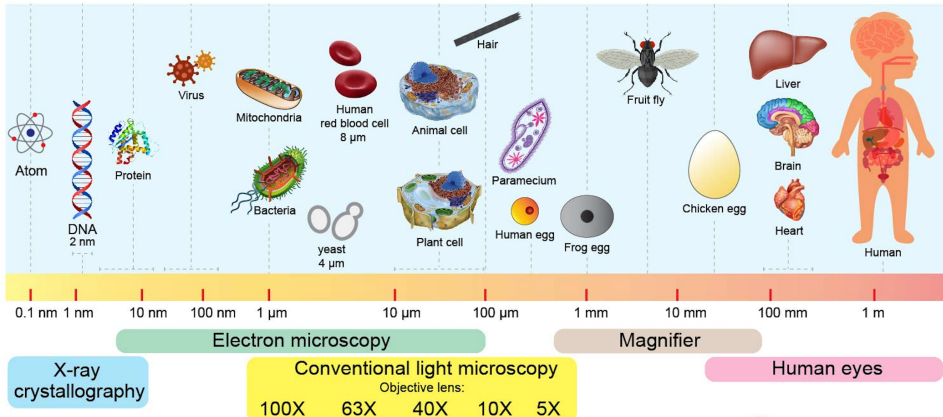
What is this talk about?

The main emphasis is to give you a short and hopefully pedestrian overview to the whys and hows of machine learning and quantum technologies, why this could (or should) be of interest and where to find info about courses and study programs at UiO.

Multiscale scientific problem, life science example

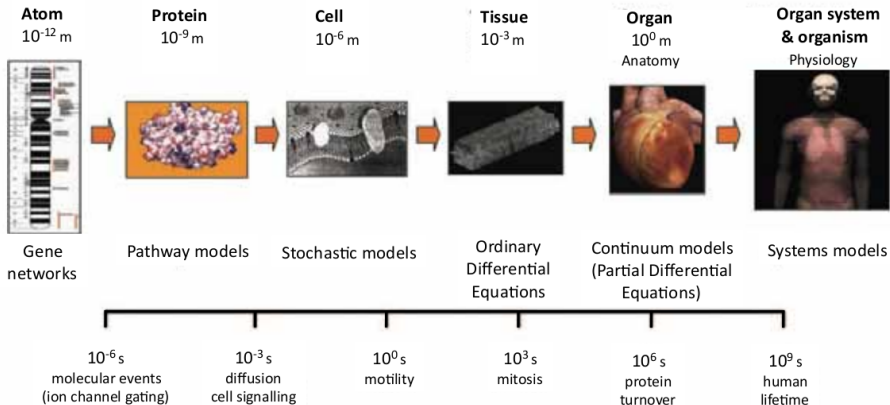
THE LENGTH SCALE OF BIOLOGY

from atoms, DNA, proteins, viruses, bacteria, mitochondria, animal and plant cells, single-celled organisms, fruit fly, organs, to our bodies

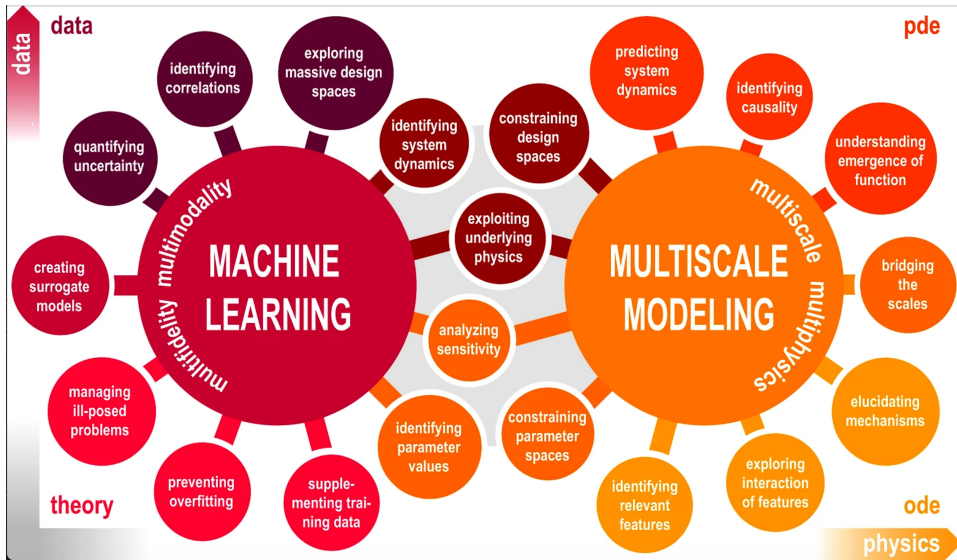


Multiscale scientific problem, life science example, with mathematical methods

The large span of spatial and temporal scales in biological systems



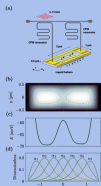
Multiscale scientific problem and Machine learning/AI



Quantum technology and machine learning/AI

Quantum Hardware

- Quantum control and sensors
- Quantum computers
- Catalysis



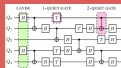
AI for Quantum Technology

- Quantum mechanical many-body theories
- Molecular dynamics
- Density functional theory

Quantum Technology and AI

Quantum Algorithms for AI

- Quantum computing
- Quantum algorithms
- Quantum machine learning
- Quantum inspired classical computing



Societal impact and products



New medicines



Better shipping



Financial innovation



Quantum readiness



Quantum sensors and computers

- Drug discovery
- Quantum sensing
- Supply chain optimization
- Financial market modelling and risk analysis
- Material discovery

What is Machine Learning?

Machine Learning (ML) is the study of algorithms that improve through data experience.

Types of Machine Learning:

- **Supervised Learning:** Labeled data for classification or regression.
- **Unsupervised Learning:** No labels; discover hidden patterns.
- **Reinforcement Learning:** Learning through interaction with the environment.

ML Workflow:

Data → Model Training → Prediction

AI/ML and some statements you may have heard (and what do they mean?)

- ① Fei-Fei Li on ImageNet: **map out the entire world of objects** (The data that transformed AI research)
- ② Russell and Norvig in their popular textbook: **relevant to any intellectual task; it is truly a universal field** (Artificial Intelligence, A modern approach)
- ③ Woody Bledsoe puts it more bluntly: **in the long run, AI is the only science** (quoted in Pamilla McCorduck, Machines who think)

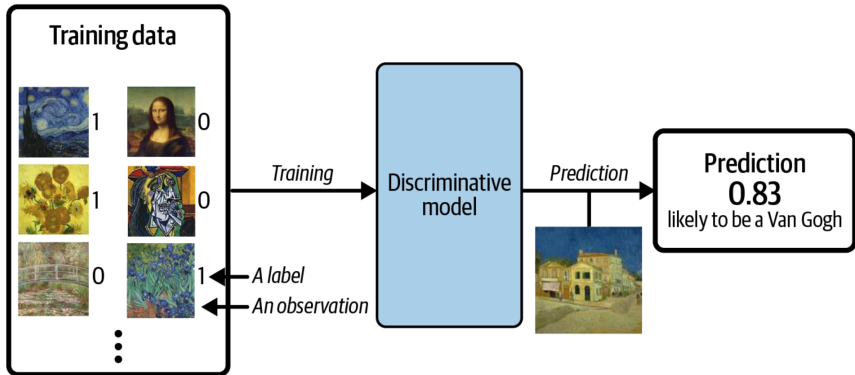
If you wish to have a critical read on AI/ML from a societal point of view, see Kate Crawford's recent text Atlas of AI.

With AI/ML we normally intend a collection of machine learning methods with an emphasis on statistical learning and data analysis

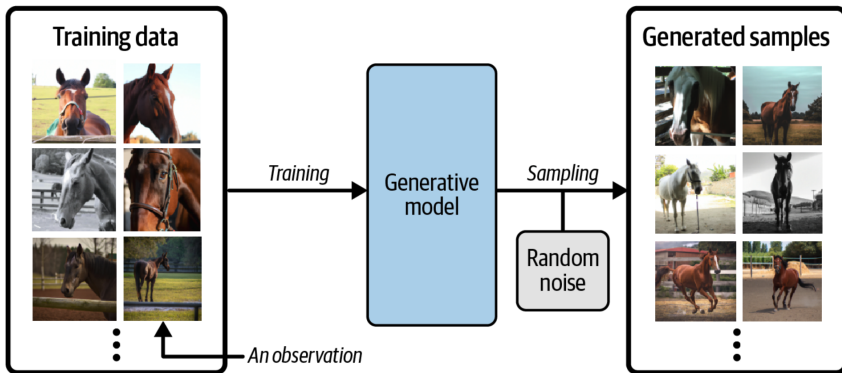
The plethora of machine learning algorithms/methods

- ① Deep learning: Neural Networks (NN), Convolutional NN, Recurrent NN, Boltzmann machines, autoencoders and variational autoencoders and generative adversarial networks, stable diffusion and many more generative models
- ② Bayesian statistics and Bayesian Machine Learning, Bayesian experimental design, Bayesian Regression models, Bayesian neural networks, Gaussian processes and much more
- ③ Dimensionality reduction (Principal component analysis), Clustering Methods and more
- ④ Ensemble Methods, Random forests, bagging and voting methods, gradient boosting approaches
- ⑤ Linear and logistic regression, Kernel methods, support vector machines and more
- ⑥ Transformers; Reinforcement Learning; Agentic AI, and more

Example of discriminative modeling, taken from Generative Deep Learning by David Foster



Example of generative modeling, taken from Generative Deep Learning by David Foster



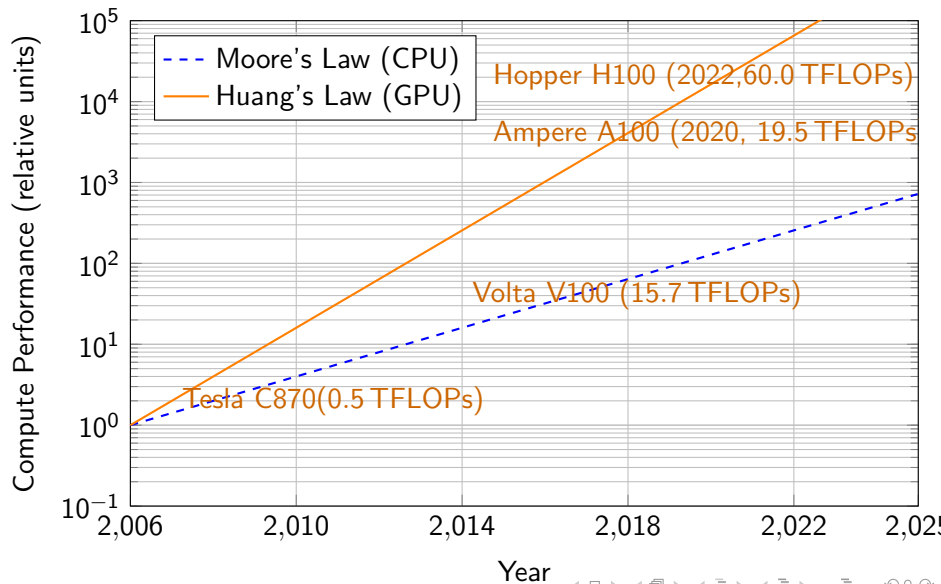
What are the basic Machine Learning ingredients?

Almost every problem in ML and data science starts with the same ingredients:

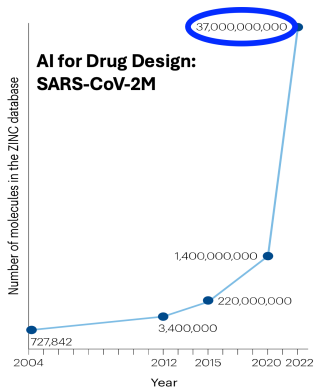
- The dataset $\mathbf{x}$ (could be some observable quantity of the system we are studying)
- A model which is a function of a set of parameters α that relates to the dataset, say a likelihood function $p(\mathbf{x}|\alpha)$ or just a simple model $f(\alpha)$
- A so-called **loss/cost/risk** function $\mathcal{C}(\mathbf{x}, f(\alpha))$ which allows us to decide how well our model represents the dataset.

We seek to minimize the function $\mathcal{C}(\mathbf{x}, f(\alpha))$ by finding the parameter values which minimize $\mathcal{C}$. This leads to various minimization algorithms. It may surprise many, but at the heart of all machine learning algorithms there is an optimization problem.

From Moore's law to Huang's law, the race for more performance



Machine learning and AI models are computationally expensive



Simulations took:

90 days on 250 GPUs & 640 cpu-core

1 day on 27000 GPU Summit supercomputer

[Nature Reviews Drug Discovery](#) **23**, 141–155 (2024)

TOP4 Powerful Supercomputers as of Nov 2024

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TDOSS, HPE DOE/NNSA/LLNL United States US\$600 million	11,039,616	1,742.00	2,746.38	29,581

Advance nuclear weapon science and scientific discovery.

nature reviews drug discovery ⁷⁶

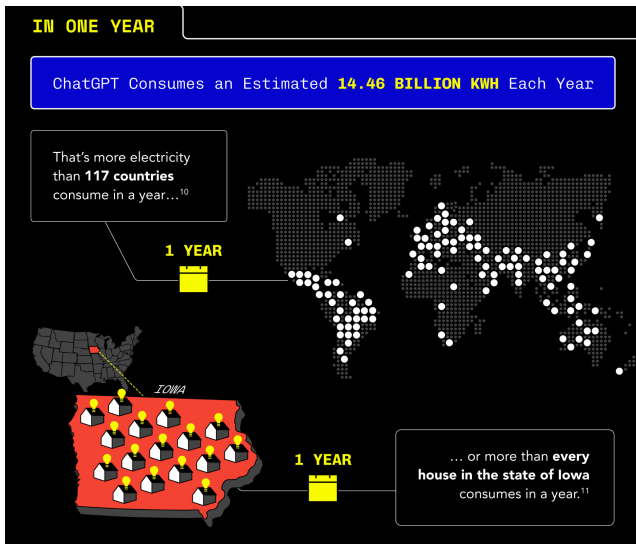
Perspective | [Published: 08 December 2023](#)

Integrating QSAR modelling and deep learning in drug discovery: the emergence of deep QSAR

Max, Slingshot-11, Intel
DOE/SC/Argonne National Laboratory
United States

4	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	4
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And power greedy, perhaps quantum computers can reduce the impact?



Taken from <https://www.businessenergyuk.com/knowledge-hub/>

And perhaps not environmentally friendly

- In Ireland, data centers consume more than 20 percent of the country's electricity.
- In Chile, precious aquifers are in danger of depletion.
- In South Africa, where blackouts have long been routine, data centers are further taxing the national grid.
- Similar concerns have surfaced in Brazil, Britain, India, Malaysia, the Netherlands, Singapore and Spain and other countries.

The future is here, sooner than we may think

Our Quantum Echoes algorithm is a big step toward real-world applications for quantum computing

Oct 22, 2025
5 min read

Our Willow quantum chip demonstrates the first-ever algorithm to achieve verifiable quantum advantage on hardware.



Hartmut Neven
Founder and Lead,
Google Quantum AI



Vadim Smelyanskiy
Director, Quantum
Pathfinding, Google
Quantum AI

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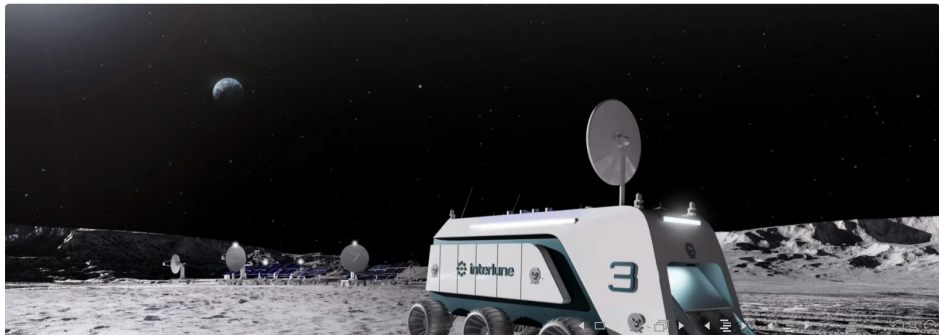


Modern quantum platforms require cooling down to extremely low temperatures

Bluefors to Source Helium-3 from the Moon with Interlune to Power Next Phase of Quantum Industry Growth

3 min read

September 16, 2025



What is quantum technology? Three main fields

Quantum computing

is a new computing paradigm that capitalizes on the laws of quantum mechanics to provide significant performance improvement for certain applications, and to enable new territories of computing beyond existing classical computing.

Quantum communication

is the secure transfer of quantum information across distances and could ensure security of communication even in the face of unlimited quantum computing power.

Quantum sensing

includes a new generation of sensors, based on quantum systems, that provide measurements of various quantities (for example, electromagnetic fields, gravity, or time) and that are orders of magnitude more sensitive than classical sensors.

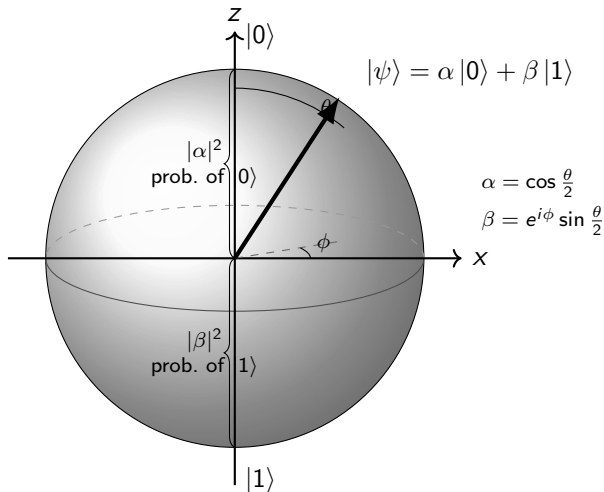
The basic concepts

Quantum technologies leverage principles of quantum mechanics to perform computations beyond classical capabilities.

Key Concepts:

- **Superposition:** Qubits can exist in a combination of states.
- **Entanglement:** Correlation between qubits regardless of distance.
- **Quantum Interference:** Probability amplitudes interfere. Is a core tool in quantum algorithms. It is used to amplify the probability of finding the correct answer while suppressing incorrect ones.

Superposition



Quantum superposition of a single qubit:
the same physical state $|\psi\rangle$ is a coherent combination of $|0\rangle$ and $|1\rangle$.

What is Quantum Entanglement?

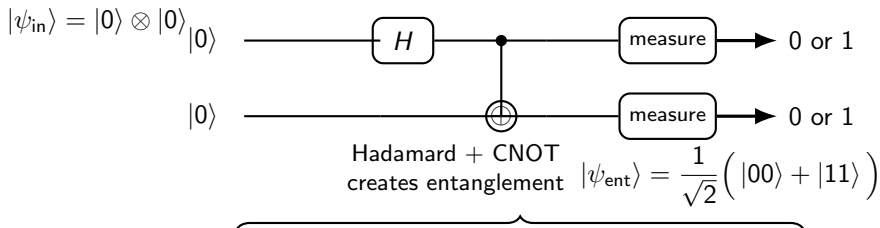
Quantum Entanglement is a quantum phenomenon where two or more particles become correlated in such a way that the state of one particle directly affects the state of the other, regardless of distance. Fundamental for making quantum gates, quantum sensors and much more.

Key Features:

- Non-local correlations
- No classical analog

Entanglement

Outcomes are
perfectly correlated:
if qubit A = 0, qubit B = 0;
if qubit A = 1, qubit B = 1.



Entanglement: the two-qubit state cannot be written as $|\phi\rangle_A \otimes |\chi\rangle_B$.
Measurement on one instantly fixes the other's outcome.

Quantum Speedups

Why Quantum?

- **Quantum Parallelism:** Process multiple states simultaneously.
- **Quantum Entanglement:** Correlated states for richer information.
- **Quantum Interference:** Constructive and destructive interference to enhance solutions, is a core tool in quantum algorithms.

Example - Grover's Algorithm:

Quantum Search Complexity: $O(\sqrt{N})$ vs. $O(N)$

Advantage: - Speedups in high-dimensional optimization and linear algebra problems.

1. Quantum Communication

Quantum Teleportation:

- Entanglement enables the transmission of quantum states using classical communication.
- No need to send the physical quantum particle.

Advantage:

- Instantaneous state transfer within quantum mechanics constraints.
- Quantum networks rely on entanglement for secure communication.

2. Quantum Cryptography

Quantum Key Distribution:

- Entanglement ensures secure communication.
- Eavesdropping disturbs quantum states, revealing interception attempts.
- Any measurement by a third party collapses the wavefunction.
- Ensures security based on quantum mechanics, not computational hardness.

Advantage: Unconditional security guaranteed by the laws of physics.

3. Quantum Computing

Speedup in Quantum Algorithms:

- Entanglement provides exponential state space.
- Quantum parallelism arises from entangled qubits.

Grover's Algorithm:

$$\mathcal{O}(\sqrt{N}) \text{ vs. } \mathcal{O}(N)$$

Shor's Algorithm:

$$\text{Factoring in } \mathcal{O}((\log N)^3)$$

4. Quantum Metrology

Quantum Metrology:

- Uses entangled states for ultra-precise measurements.
- Overcomes the classical shot-noise limit.

Heisenberg Limit:

$$\Delta\theta \geq \frac{1}{N},$$

where N is the number of entangled particles.

Advantage:

- Quantum entanglement improves sensitivity beyond classical limits.

Challenges of Quantum Entanglement

Decoherence:

- Entangled states are fragile.
- Interaction with the environment collapses the wavefunction.

Scalability:

- Difficult to entangle large numbers of qubits.
- Error correction requires complex protocols.

Measurement Problem:

- Measurement destroys entanglement.
- Trade-off between information gain and entanglement preservation.

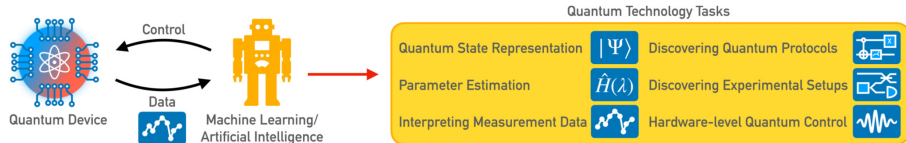
What is Quantum Machine Learning?

Quantum Machine Learning (QML) integrates quantum computing with machine learning algorithms to exploit quantum advantages.

Motivation:

- High-dimensional Hilbert spaces for better feature representation.
- Quantum parallelism for faster computation.
- Quantum entanglement for richer data encoding.

Quantum technologies and ML/AI



Can I study these topics at UiO?

- Several courses on Machine learning and AI at UiO, from bachelor level to PhD level, see <https://github.com/CompPhysics/MachineLearning>
- Quantum technologies, quantum information theory, quantum computing, quantum sensing and more.
 - ① Quantum Technology Bachelor of Science program <https://www.uio.no/studier/program/kvanteteknologi/>
 - ② Quantum Technology Master of Science program <https://www.uio.no/english/studies/programmes/quantum-technology/>
 - ③ Computational Science Master of Science program <https://www.uio.no/english/studies/programmes/computational-science-master/programme-options/quantum-technology/index.html>